Experiment - 1

AIM:-

To design an Entity-Relationship (ER) diagram for an Indian e-commerce platform consisting of Customer, product, order, Order Item, Seller, category, Payment, Delivery and address, and to identify primary keys, composite and multi-valued attributes, weak entities, specialization / generalization, and participation constraints

→ ER Diagram Components.

Entities:-

- Customer
- Product
- Order
- Order Item
- Seller
- Category
- Payment
- Delivery
- Address.

→ Primary keys.

- Customer → Customer ID
- Product → Product-ID
- Order → Order-ID
- Order Item → (Order-ID, Item NO) [Composite primary key]
- Seller → Seller-ID
- Category → Category-ID
- Payment → Payment-ID

→ Relationships and cardinality

1. Customer places Order → 1:N
2. Customer has address → 1:N
3. Order Contains Order Item → 1:N
4. Product appears in Order Item → 1:N
5. Seller Sells Products → 1:N
6. Category contains Product → 1:N
7. Order has payment → 1:1
8. Order has Delivery → 1:1

Participation Constraints.

- Customer-Order : Order has total participation.
- Order-Order Item : Both Order and Order Item have total participation.
- Product-Order Item : Order Item has total participation.
- Seller-Product : Product has total participation.
- Category-Product : Product has total participation.
- Order-Payment. Payment has total participation.
- Order-Delivery : Delivery has total participation.

- Delivery → Delivery_ID
- Address → Address_ID

→ Important Attributes.

1. Customer
   - Customer_ID
   - Name
   - Phone

2. Product
   - Product_ID
   - Product_Name
   - Price

3. Order
   - Order_ID
   - Order_Date
   - Total_Amount
   - Order_Status

4. Order Item
   - Item_No
   - Quantity
   - Unit_Price
   - Discount

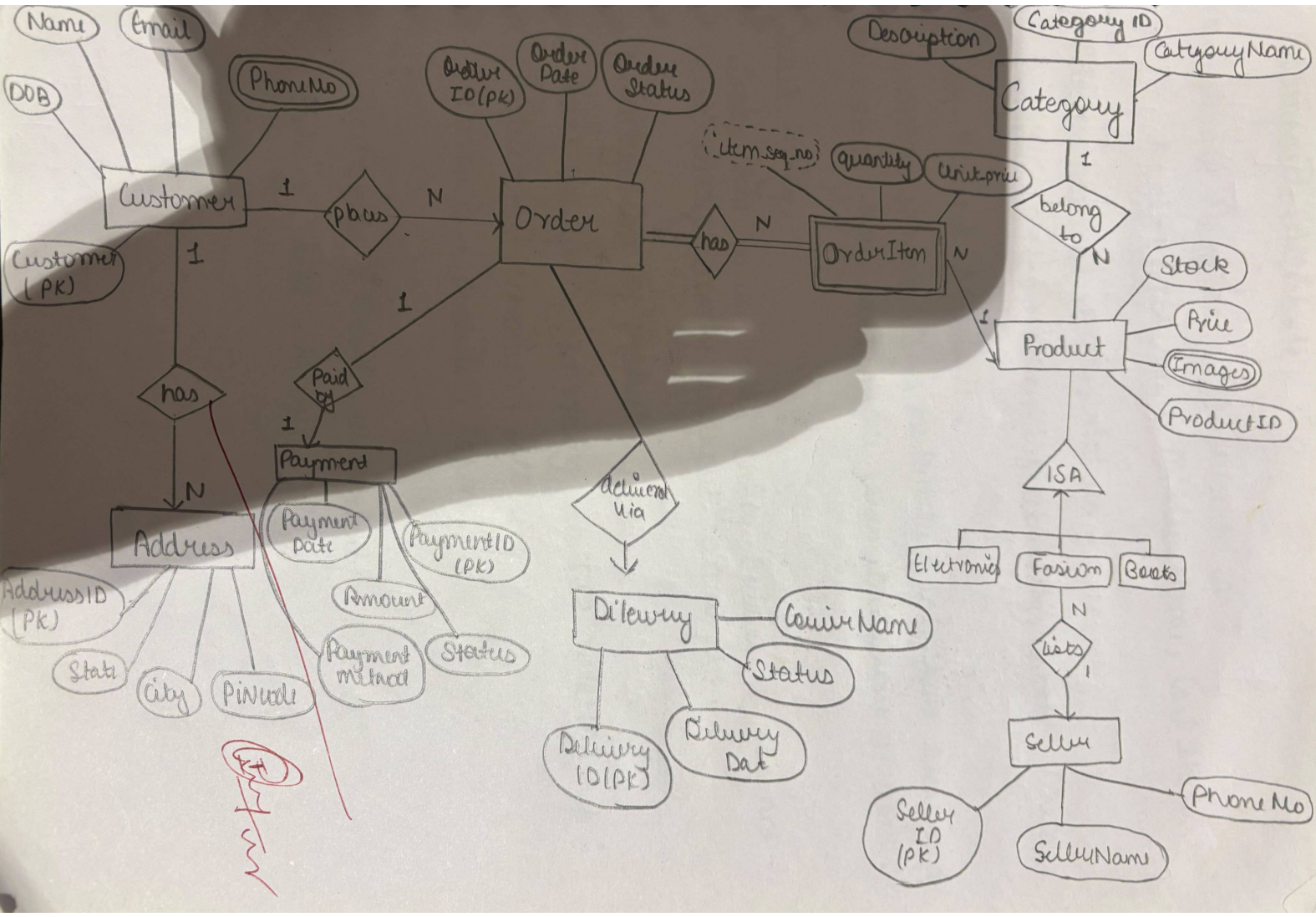
5. Seller.
   - Seller_ID
   - Seller_Name
   - Phone

6. Category
   - Category_ID
   - Category_Name
   - Description

7. Payment
   - Payment_ID
   - Payment_Date
   - Amount.
   - Transaction_ID.
   - Payment_Method.

8. Delivery
   - Delivery_ID
   - Delivery_Date
   - Status
   - ~~Charges~~

9. Address.
   - Address_ID
   - House_No
   - Street.
   - City
   - State
   - Pincode.


Name
Email
DOB
PhoneNo
Customer
Customer (PK)
1
places
N
Order
Order ID (PK)
Order Date
Order Status
has
N
OrderItem
item_seq_no
quantity
Unit_price
N
1
Description
Category ID
CategoryName
Category
1
belong to
N
Stock
Price
Images
ProductID
Product
1
has
1
N
Paid by
1
Payment
Address
Payment Date
PaymentID (PK)
Amount
Payment Method
Status
AddressID (PK)
State
City
PINcode
Delivered via
Delivery
CourierName
Status
Delivery ID(PK)
Delivery Date
ISA
Electronics
Fashion
Beauty
N
lists
1
Seller
Seller ID (PK)
SellerName
PhoneNo

Composite Attribute.

- Customer_Name can be represented as:
  Name → First_Name, Middle_Name, Last_Name.

- Address can be represented as:
  Address → House_No, street, city, state, Pincode.

Multi-Valued Attribute.

customer_Phone is a multivalued attribute because a customer may have more than one phone number.

Weak Entity.

Order Item is a weak entity because is can not exist independently without an order.

Specialization of product.

Product is specialized using an ISA relationship:

Product → Electronics, clothing
Electronics → mobile, Laptop.

Eg:

- Electronics: Brand, Warranty_Period.
- Clothing: Size, Fabric, Color.

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